

Aryan Hemendra Shah
University of Miami
Miami, FL 33146
ahs222@miami.edu

June 4, 2026

The Editors
Life Sciences in Space Research
Elsevier

Dear Editors,

I am submitting for your consideration the manuscript “**An open-source GCR dosimetry pipeline for Mars mission risk assessment: organ-specific REID, SEP acute hazard, and dual-use proton therapy RBE benchmarking**” for publication in *Life Sciences in Space Research*.

What the paper does. The manuscript presents an open-source Python package (GCR-DOSIMETRY-PIPELINE) that integrates GCR spectrum generation, CSDA slab transport, organ-specific cancer risk (REID), Solar Energetic Particle acute dosimetry, and global uncertainty quantification via Latin Hypercube Sampling into a single reproducible workflow with 82 automated tests and a public repository.

What is new. I am transparent that the individual components — force-field modulation, CSDA transport, REID estimation — are established methods. The contributions are threefold.

First, no currently available open tool combines organ-specific REID, SEP acute risk, and global uncertainty propagation in a single workflow reproducible from a standard `pip install`.

Second, the per-species LET decomposition shows that carbon ions account for approximately 59% of unshielded GCR absorbed dose at a field LET of $18.7 \text{ keV}/\mu\text{m}$ — the same LET range that governs Bragg-peak RBE uncertainty in clinical proton therapy. The pipeline computes Wedenberg and McNamara RBE for both GCR and clinical proton beam geometries from one codebase, enabling direct comparison of quality factors across both fields. To my knowledge this connection has not been demonstrated quantitatively in an open tool.

Third, the Latin Hypercube Sampling ensemble shows that DDREF and excess risk scaling together account for more than 70% of REID output variance, quantifying why physics improvements alone cannot close mission risk uncertainty. This is a directly actionable finding for research prioritization.

Scope and limitations. The transport model uses CSDA with exponential nuclear attenuation and does not capture heavy-ion fragmentation secondaries. This limitation is stated explicitly and its domain of validity is discussed. The per-species calibration to MSL/RAD is

described as calibration, not independent prediction, and the validation evidence is presented accordingly. The REID framework follows Cucinotta (2013); the 2023 NASA update is noted as a limitation for future work.

Relevance to *Life Sciences in Space Research*. This manuscript is squarely within the journal’s scope: human spaceflight radiation risk, Mars mission planning, and space radiation biology. The dual-use LET framing (space radiation $\leftrightarrow$ proton therapy) additionally connects to an emerging readership interested in high-LET radiobiology across both domains.

Data and code availability. All source code, input data, validation scripts, and TOPAS benchmark geometries are openly available on GitHub at the repository [aryanhshah8/gcr-dosimetry-pipeline](https://github.com/aryanhshah8/gcr-dosimetry-pipeline), and no proprietary datasets are required to reproduce any results.

I confirm that this manuscript has not been submitted elsewhere, that I am the sole author, and that I have no conflicts of interest to declare. I look forward to the reviewers’ comments.

Sincerely,

Aryan Hemendra Shah
University of Miami
ahs222@miami.edu
ORCID: 0009-0008-0518-428X